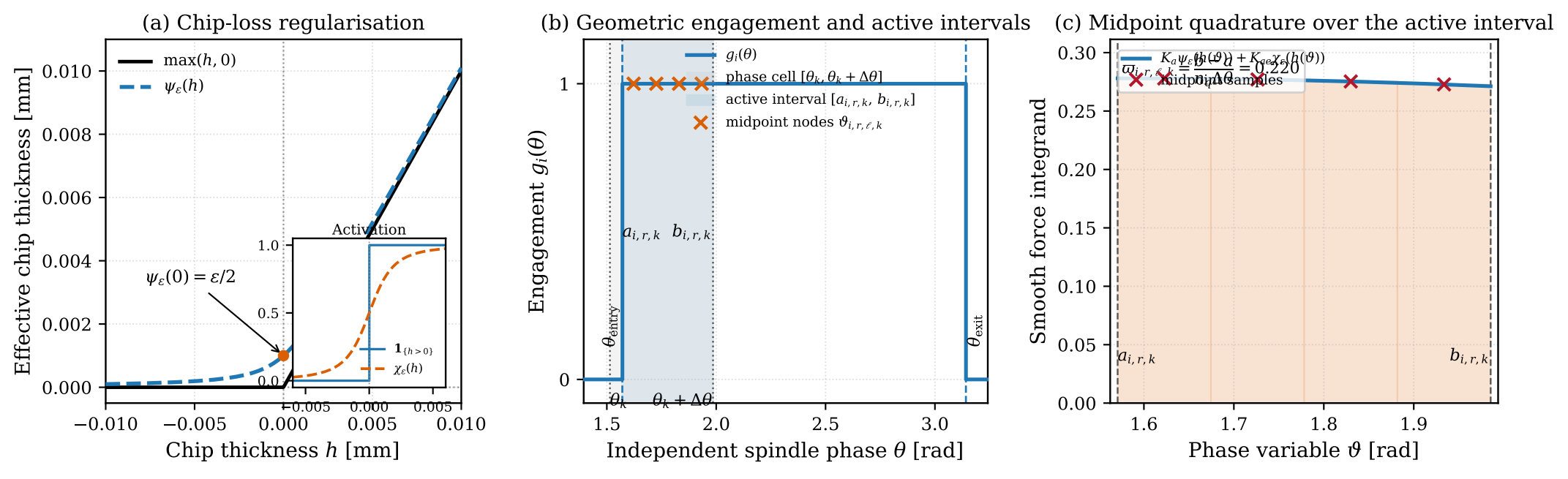
(a) Chip-loss regularisation
max(h, 0)
ψε(h)
ψε(0) = ε/2
Activation
1.0
0.5
0.0
−0.005
0.000
0.005
1{h > 0}
χε(h)
Effective chip thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]
(b) Geometric engagement and active intervals
gi(θ)
phase cell [θk, θk + Δθ]
active interval [ai, r, k, bi, r, k]
midpoint nodes ϑi, r, ℓ, k
Engagement gi(θ)
1
0
ai, r, k
bi, r, k
θentry
θexit
θk
θk + Δθ
1.5
2.0
2.5
3.0
Independent spindle phase θ [rad]
(c) Midpoint quadrature over the active interval
Kaψε(h(ϑ)) + Kaeχε(h(ϑ))
ωi, r, ℓ, k = Δθ/Q = 0.220
midpoint samples
Smooth force integrand
0.30
0.25
0.20
0.15
0.10
0.05
0.00
ai, r, k
bi, r, k
1.6
1.7
1.8
1.9
Phase variable ϑ [rad]